

A void growth and coalescence dependent anisotropic damage model for polymeric foams

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Abstract

We present an elastic-damage constitutive model for polymeric foam based on thermodynamics framework to consider the effects of anisotropy and the growth and coalescence of cavities. The evolution equation of the proposed model describes the material behavior sustaining anisotropic and unilateral damage. To carry out finite element analysis, the material properties for various polymeric foams are applied to the proposed damage model; we thus implement the proposed damage model in the commercial finite element program as a user-defined material subroutine. In order to validate the proposed anisotropic damage model, the numerical results are compared to the results of a series of tensile and compressive tests on various polymeric foams. Additionally, we propose the failure criterion for this damage model as based on the mass-dependent energy per unit mass at failure. The developed damage model can be used for further research on damage mechanics and finite element analysis of polymeric foams in continuum mechanics.

Keywords: Polymeric foam, Porosity/Voids, Damage mechanics, Anisotropic damage, Finite element analysis (FEA), Continuum mechanics

1. Introduction

Polymeric foams are widely employed throughout various industries, as they possess numerous favorable properties, such as low density, low thermal conductivity, low water absorption and permeability, impact absorption, and dimensional stability [1–2]. The liquefied natural gas (LNG) insulation system implemented in LNG carriers is a notable example of polymeric foam use. For the LNG insulation system, a significant load component is the sloshing impact load, which generates a shock wave. Although the principal load in an LNG insulation system results from compressive loading, shock waves may induce tensile stress [3–4]. Therefore, polymeric foams are subjected to both compressive and tensile loading. When a compressive load is applied to polymeric foams, the ductile behavior of the material appears, whereas for some polymeric foams, such as PUF [5–6], R-PUF [7], and polyvinyl chloride (PVC) [6, 8], the brittle failure behaviors are observed under tensile loading.

To date, researchers have performed many experiments and numerical studies that focused on evaluating the structural performance and behavior of polymeric foams. Saint-Michel et al. [9–10] evaluated the mechanical properties and behavior of polyurethane foam (PUF) with various densities and filler sizes by conducting compressive and tensile tests. Park et al. [11] experimentally investigated the mechanical properties and fracture behavior of polymeric foams at various strain rates and temperatures by performing a series of compression tests. Additionally, tensile and fracture-toughness tests were conducted for glass fiber-reinforced polyurethane foam (R-PUF) to analyze the failure mechanisms [12]. In Ref. [5], the mechanical impact of adding varying mass fractions of milled glass fibers and nanoparticles to low-density PUF is investigated via tensile, compressive, and shear testing. Andersons et al. [13] evaluated the anisotropy of the stiffness and strength of rigid low-density closed-cell polyisocyanurate foam (PIR). Lee et al. [14] described the hardening and softening phenomena of R-PUF under compressive loading via a Frank-Brockman-type isotropic elasto-viscoplastic model implementing various temperatures and strain rates. Lee et al. [15] described the nonlinear behavior of PUF by considering the density effect in their implementation of a Gurson model with an acceleration factor. Authors proposed a unified anisotropic elasto-viscoplastic damage model and computational analysis method for R-PUF [7]; they also evaluated the inelastic constitutive behavior and debonding failure of a

polymer-type adhesively bonded joint using a computational approach [16]. Many constitutive and damage models have been proposed to evaluate the compressive behavior of polymeric foams. In addition, Dillard et al. [17] proposed an anisotropic damage model for ductile foams in the framework of the micromorphic continuum theory. In Ref. [18], a model coupling non-linear elasticity, viscosity, and damage was introduced to describe the response subjected to uniaxial cyclic compression for a polyurethane foam. However, although it is important to describe the ductile behavior in order to evaluate the cracks occurring in composite structures subjected to extreme impact loading, there are few studies that have proposed anisotropic brittle damage models for polymeric foams subjected to tensile loading.

Generally, there are many brittle damage theories that were derived as based on initial microdefects or a deterministic approach [19–27]. In these damage models, the damage state can be represented by implementing a damage tensor and/or damage-strengthening internal variable, and an internal variable to represent the slipped state. To describe the material behavior of polymeric foams, the use of these damage models enables damage model prediction that is in good agreement with the tensile behavior. However, a disadvantage of these conventional damage theories is that no relationship between voids or cavities and damage is considered or represented. Polymeric foams are porous materials (unlike typical brittle materials, such as concrete, rock, and ceramics) and have spherical gas bubbles, i.e., voids [15], which arise from the mixing and foaming process, and brittle cracks often begin at a pre-existing defect, such as a void or inclusion [28]. The voids in the polymeric foams grow as a tensile or compressive load is applied to the material in elastic and plastic regimes [29–30]. Therefore, a brittle damage model for the polymeric foam should incorporate physical representations of the voids effects since the volume fraction of voids is related to the damage state [31–33].

There are various ductile damage models that are focused on modeling voids in materials [31–35], such as the Rousselier model, Gurson model, and modified Gurson model. The Rousselier model describes the process of nucleation, growth, and coalescence of spherical voids [32], and the Gurson model yields a constitutive model of the nucleation and growth of voids due to plastic deformation [32–34]. Subsequently, Tvergaard and Needleman [35] proposed a modified Gurson model (GTN model) to incorporate the effects of void coalescence. Although each of these damage models effectively achieves their purpose,

they are all elastic-plastic-ductile damage models; as such, there remains no brittle damage model for the growth and coalescence of voids in materials.

Thus, in this study, a novel brittle damage model describing the growth and coalescence of voids is developed within the framework of continuum damage mechanics and as based on irreversible thermodynamics. To consider the effects of anisotropy in materials, an anisotropic brittle damage model is proposed. The developed anisotropic brittle damage model is explicitly formulated and implemented as a user-defined material subroutine (UMAT) in the commercial finite element program ABAQUS for computation. Subsequently, numerical simulations are performed to validate the proposed damage model, then we compare the results of tensile and compressive tests. Finally, upon evaluation of the energy per unit mass at the time of failure for various polymeric foams, for the proposed damage model we suggest a failure criterion as an empirical formula determined via regression analysis.

2. Void growth and coalescence dependent Brittle damage model

2.1 Free energy function and damage surface

In prior research, several constitutive equations have been formulated in order to describe anisotropic damage effects, as the macroscopically observed degradation of the mechanical properties of engineering materials is anisotropic [36]. For instance, in Refs. [37–38], the brittle damage is described by the compliance tensor. In addition, a symmetric second-order tensor $\mathbf{D}$ can be used to describe anisotropic material degradation [39–40]. Moreover, employing a fourth-order damage tensor $\mathbb{D}$ is the mathematical model for anisotropic damage. Although the second-order damage tensor remains the most popular owing to its simplicity in terms of anisotropic damage, the advantage of the fourth-order tensor is that it can change every component of the stiffness tensor [41–42].

In this study, to describe the degradation of the elastic stiffness, we employ the fourth-order elastic modulus tensor $\mathbb{C}$ as the damage variable of the proposed brittle damage model, rather than the fourth-order damage tensor $\mathbb{D}$, since the damaged stiffness $\mathbb{C}$ possesses a one-to-one

correspondence with the damage tensor $\mathbb{D}$ [25].

We assume that the free energy function $\psi = \psi^E + \psi^D$ describes the free energy caused by elastic deformation ψ^E , and the energy related to damage ψ^D , where the functions $\rho\psi^E(\boldsymbol{\varepsilon}, \mathbb{C}) = \frac{1}{2}\boldsymbol{\varepsilon}:\mathbb{C}:\boldsymbol{\varepsilon}$ and $\psi^D = \psi^D(\beta)$ are derived as based on the damage theory proposed by Ju [25], and the Rousselier model [31], respectively; ρ , $\boldsymbol{\varepsilon}$, and β are the mass density, elastic strain tensor, and damage-strengthening internal variable, respectively; the symbol $(:)$ denotes the tensorial product contracted on two indices.

Using the Clausius-Duhem inequality [43–45] for an isothermal process of a brittle material yields the following constitutive equations: $\boldsymbol{\sigma} = \mathbb{C}:\boldsymbol{\varepsilon}$. Furthermore, the Clausius-Duhem inequality eventually reduces to

$$\Phi = \mathbf{X} \cdot \mathbf{J} \geq 0 \quad (1)$$

where $\mathbf{X} = (\mathbb{Y}/\rho, B)$ and $\mathbf{J} = (\dot{\mathbb{C}}, \dot{\beta})$ when $\mathbb{Y} = -\rho \frac{\partial \psi^E}{\partial \mathbb{C}}$ and $B = -\rho \frac{\partial \psi^D}{\partial \beta}$. The symbol Φ denotes the dissipation per unit volume of the material. $\mathbf{X}$ and $\mathbf{J}$ denote the generalized force vector and associated generalized flux vector, respectively. The evolution equations of $\mathbf{J}$ can be derived as based on a damage loading surface (or dissipation potential function) defined as a function of $\mathbf{X}$, since the dissipation per unit volume Φ is expressed in the form of Eq. (1) [45–48].

Ju [25] assumed that the damage develops only when the damage-associated variable $\mathbb{Y}$ remains on the loading surface. Hayakawa and Murakami [49] proposed a damage surface $F^D = 0$ using the variable B that is a conjugate force of internal state variable β . However, it should be noted that the damage models proposed by Ju [25] and Hayakawa and Murakami [49] do not require a hypothesis relating to the growth and coalescence of voids. In contrast, in order to describe the process of nucleation, growth, and coalescence of spherical cavities, Rousselier [31] used a relationship between the variable $B(\beta)$ and the associated variable related to the hydrostatic stress and mass density $h(\sigma_H/\rho)$. Similarly, we use the relationship to take account of the growth of voids. Therefore, the geometry of the damage surface is dependent not only on the damage-associated variable $\mathbb{Y}$ and variable B , but also on the ratio

of the hydrostatic stress to mass density σ_H/ρ . Accordingly, the damage potential may be assumed as follows:

$$F^D(\mathbb{Y}/\rho, \beta, \sigma_H/\rho) = g(\mathbb{Y}/\rho) + B(\beta)h(\sigma_H/\rho) \quad (2)$$

such that the generalized flux $\mathbf{J}$ can be derived from F^D as is given below.

$$\mathbf{J} = \dot{\Lambda}^D \frac{\partial F^D}{\partial \mathbf{x}} \quad (3)$$

where $\dot{\Lambda}^D$ is a damage multiplier. Substitution of the dissipation potentials of Eq. (2) into Eq. (3) provides the following equations:

$$\dot{\mathbb{C}} = \dot{\Lambda}^D \frac{\partial F^D}{\partial (\mathbb{Y}/\rho)} \quad (4)$$

$$\dot{\beta} = \dot{\Lambda}^D \frac{\partial F^D}{\partial B} = \dot{\Lambda}^D h(\sigma_H/\rho) \quad (5)$$

The plasticity models of spherical voids growth showed that the dissipation potential function is dependent on the hydrostatic stress and strain [31–32, 35]. The Rousselier model, one of the plasticity models, employed $\dot{\varepsilon}_H^p = \frac{1}{3} \text{tr} \left[\dot{\Lambda}^D \frac{\partial F^D}{\partial (\sigma/\rho)} \right]$ as an evolution equation of damage development, where $\dot{\varepsilon}_H^p$ is the hydrostatic plastic strain rate. Similarly, in order to describe the process of growth and coalescence of spherical cavities in the polymeric foams, we postulate the following relationship because the cavities in the polymeric foams grow under tensile and compressive loading in the elastic regime [29–30]:

$$\dot{\varepsilon}_H = \frac{1}{3} \text{tr} \left[\dot{\Lambda}^D \frac{\partial F^D}{\partial (\sigma/\rho)} \right] = \frac{1}{3} \dot{\Lambda}^D B(\beta) h'(\sigma_H/\rho) \quad (6)$$

where σ_H and ε_H are the hydrostatic stress ($= \sigma_{kk}/3$) and hydrostatic strain, respectively. The loading-unloading condition for the damage evolution is specified as follows:

$$\dot{\Lambda}^D > 0 \Rightarrow F^D = 0 \quad (7)$$

2.2 Influence of unilateral effect on damage

Murakami [50] introduced a scalar variable ξ for the use instead of the damage-associated variable $\mathbb{Y}$ to represent the unilateral effect (due to the opening-closure of microcracks) on damage development.

Similarly, we introduce a new damage value ζ for the use in place of $\mathbb{Y}/\rho$ for representing the mass density and unilateral effect on damage development:

$$\zeta \equiv \text{tr}(\mathbb{Y}/\rho : \mathbb{C}_{AC}) \quad (8a)$$

$$\mathbb{C}_{AC} = \mathbb{P}_{\varepsilon}^+ : \mathbb{C}_0 : \mathbb{P}_{\varepsilon}^+ \quad (8b)$$

$$\mathbb{P}_{\varepsilon}^+ \equiv \sum_{i=1}^3 H(\varepsilon_i) \mathbf{n}_i^{(\varepsilon)} \otimes \mathbf{n}_i^{(\varepsilon)} \otimes \mathbf{n}_i^{(\varepsilon)} \otimes \mathbf{n}_i^{(\varepsilon)} \quad (8c)$$

where $\text{tr}(\cdot)$ is the trace of the tensor (here, the fourth-order tensor $\mathbb{Y}/\rho : \mathbb{C}_{AC}$ can be represented in the form of a six-by-six matrix using Voigt notation). $\mathbb{C}_{AC}$ is the activated elastic modulus tensor. $\mathbb{P}_{\varepsilon}^+$ is a positive orthogonal projection tensor of the strain tensor $\boldsymbol{\varepsilon}$ with respect to its principal directions $\mathbf{n}_i^{(\varepsilon)}$; $H(\cdot)$ is the Heaviside function. ε_i is the i -th strain tensor component in the Voigt notation.

Eq. (8) gives

$$\frac{\partial \zeta}{\partial (\mathbb{Y}/\rho)} = \mathbb{C}_{AC} = \mathbb{P}_{\varepsilon}^+ : \mathbb{C}_0 : \mathbb{P}_{\varepsilon}^+ \quad (9)$$

By using the variable ζ of Eq. (8) instead of the damage-associated variable $\mathbb{Y}/\rho$, Eq. (2) can be expressed in the following form:

$$\bar{\mathbb{F}}^D(\zeta, \beta, \sigma_H/\rho) = \bar{\mathbf{g}}(\zeta) + \mathbf{B}(\beta) \mathbf{h}(\sigma_H/\rho) \quad (10a)$$

$$\bar{\mathbf{g}}(\zeta) = \mathbf{g}(\mathbb{Y}/\rho) \quad (10b)$$

where $\bar{\mathbf{g}}$ is a function of the new damage value ζ to represent the unilateral effect on the damage.

With the consistency condition for the dissipation potential function, we have

$$\dot{\bar{\mathbb{F}}}^D = \frac{\partial \bar{\mathbf{g}}(\zeta)}{\partial \zeta} \dot{\zeta} + \frac{\partial \bar{\mathbb{F}}^D}{\partial \beta} \dot{\beta} + \frac{\partial \bar{\mathbb{F}}^D}{\partial (\sigma_H/\rho)} \left(\frac{\dot{\sigma}_H}{\rho} \right) = 0 \quad (11)$$

Accordingly, we define an evolution equation

$$\dot{\mathbb{C}} = \dot{\Lambda}^D \frac{\partial \bar{\mathbb{F}}^D}{\partial (\mathbb{Y}/\rho)} = \dot{\Lambda}^D \frac{\partial \bar{\mathbb{F}}^D}{\partial \zeta} \frac{\partial \zeta}{\partial (\mathbb{Y}/\rho)} = \dot{\Lambda}^D H \mathbb{C}_{AC} \quad (12a)$$

where

$$H \equiv \frac{\partial \bar{\mathbf{g}}(\zeta)}{\partial \zeta} = - \left[\frac{\partial \bar{\mathbb{F}}^D}{\partial \beta} \dot{\beta} + \frac{\partial \bar{\mathbb{F}}^D}{\partial (\sigma_H/\rho)} \left(\frac{\dot{\sigma}_H}{\rho} \right) \right] / \dot{\zeta} \quad (12b)$$

2.3 Mass conservation

To derive the specific expressions of the equations presented in Section 2.2, we need to determine the functions $\rho(\beta)$ and $h(\sigma_H/\rho)$. The law of conservation of mass is

$$\dot{\rho} + 3\rho\dot{\varepsilon}_H = 0 \quad (13a)$$

or

$$3\rho\dot{\varepsilon}_H = -\rho'(\beta)\dot{\beta} \quad (13b)$$

Substituting Eqs. (5) and (6) into Eq. (13), we obtain the following relation:

$$\frac{h'(\sigma_H/\rho)}{h(\sigma_H/\rho)} = -\frac{\rho'(\beta)}{B(\beta)\rho(\beta)} \quad (14)$$

Since the left- and right-hand sides of Eq. (14) are each functions of different variables, the relation only holds when the both hand sides have a constant of identical value. Here, this constant value is set as $1/\sigma^*$, where σ^* is an adjustable parameter, translating the resistance to the growth and coalescence of cavities [31].

By substituting Eq. (14) into Eq. (13), we can identify the relationship between the damage-strengthening internal variable and hydrostatic strain:

$$B(\beta)\dot{\beta} = 3\sigma^*\dot{\varepsilon}_H \quad (15)$$

Via integration, we obtain the following relation:

$$\int B(\beta) d\beta = \int 3\sigma^* d\varepsilon_H \quad (16a)$$

or

$$\int B(\beta) d\beta = 3\sigma^*\varepsilon_H \quad (16b)$$

In addition, we derive the relationship between the mass density and hydrostatic strain via Eq. (13a).

$$\rho(\varepsilon_H) = \rho_0 \exp(-3\varepsilon_H) \quad (17)$$

where ρ_0 is the initial mass density.

2.4 Evolution equation of the proposed damage model

The mechanical behavior of the brittle materials is illustrated schematically in Fig. 1a for tension loading. The elastic modulus rapidly decreases after the level of strain reaches threshold of strain ε' . In order to

account for the increase in the absolute value of $\dot{\mathbb{C}}$ as ε increases, we postulate the following relationship between $B(\beta)$ and β :

$$B(\beta) = C \exp(-K_d \beta) \quad (18)$$

where C and K_d are adjustable parameters.

Eq. (6) is used to calculate the damage multiplier. Then, by substituting the damage multiplier and Eq. (12b) into Eq. (12a), we derive the following relation:

$$\dot{\mathbb{C}} = -\frac{3\dot{\varepsilon}_H}{\dot{\zeta}} \left[\frac{B'(\beta)h(\sigma_H/\rho)}{B(\beta)h'(\sigma_H/\rho)} \dot{\beta} + \left(\frac{\dot{\sigma}_H \rho - \sigma_H \dot{\rho}}{\rho^2} \right) \right] \mathbb{C}_{AC} \quad (19)$$

By substituting the relationships defined in Eqs. (13)–(18) into Eq. (19), we obtain the evolution equation of the proposed model (see Eq. (20)).

$$\dot{\mathbb{C}} = -\frac{3\dot{\varepsilon}_H}{\dot{\zeta}} \left[3 \left(\sigma^{*2} \frac{B'(\beta)}{\{B(\beta)\}^2} + \frac{\sigma_H}{\rho} \right) \dot{\varepsilon}_H + \frac{\dot{\sigma}_H}{\rho} \right] \mathbb{C}_{AC} \quad (20a)$$

$$\dot{\mathbb{C}} = -\frac{3\dot{\varepsilon}_H}{\dot{\zeta}} \left[3 \left(-\frac{\sigma^{*2} K_d}{C} \exp(K_d \beta) + \frac{\sigma_H}{\rho} \right) \dot{\varepsilon}_H + \frac{\dot{\sigma}_H}{\rho} \right] \mathbb{C}_{AC} \quad (20b)$$

Substituting Eq. (18) into Eq. (16), we obtain

$$-\frac{C}{K_d} (\exp(-K_d \beta) - 1) = 3\sigma^* \varepsilon_H \quad (21a)$$

or

$$\frac{K_d}{C} \exp(K_d \beta) = \left(-3\sigma^* \varepsilon_H + \frac{C}{K_d} \right)^{-1} \quad (21c)$$

Thus, the evolution equation of the proposed model finally yields

$$\dot{\mathbb{C}} = -\frac{3\dot{\varepsilon}_H}{\dot{\zeta}} \left[3 \left(\sigma^{*2} \left(3\sigma^* \varepsilon_H - \frac{C}{K_d} \right)^{-1} + \frac{\sigma_H}{\rho} \right) \dot{\varepsilon}_H + \frac{\dot{\sigma}_H}{\rho} \right] \mathbb{C}_{AC} \quad (22a)$$

For simplification, we introduce new material constants $p = 3/\sigma^*$ and $q = C/(K_d \sigma^{*2})$. Then, Eq. (22a) is written as

$$\dot{\mathbb{C}} = M \mathbb{C}_{AC} \quad (22b)$$

where

$$M = -\frac{3\dot{\varepsilon}_H}{\dot{\zeta}} \left[3 \left((p\varepsilon_H - q)^{-1} + \frac{\sigma_H}{\rho} \right) \dot{\varepsilon}_H + \frac{\dot{\sigma}_H}{\rho} \right] \quad (22c)$$

Although M is identical along each direction, the increments of the elastic modulus tensor can differ along each direction when the initial elastic tensor $\mathbb{C}_0$ is anisotropic. However, in

this study, to change each individual component of the tensor $\mathbb{M}$, a fourth-order tensor $\mathbb{M}$ is introduced although its usage can increase the complexity of the model. Then, we obtain

$$\dot{\mathbb{C}} = \mathbb{M} : \mathbb{C}_{AC} \quad (23)$$

There are numerous possibilities in terms of anisotropic effect tensor $\mathbb{M}$ configurations. In terms of the damage effect tensor, a simple form can be derived by introducing material damage in only the principal direction [51–54]. Similarly, in this study, we introduce p and q in only the principal directions of the strain, since $\dot{\mathbb{C}}$ is dependent on p and q .

If the following Voigt notations are used, the stress and strain tensors are represented as follows:

$$\boldsymbol{\sigma} = \{\sigma_1 \quad \sigma_2 \quad \sigma_3 \quad \sigma_4 \quad \sigma_5 \quad \sigma_6\}^T = \{\sigma_{11} \quad \sigma_{22} \quad \sigma_{33} \quad \sigma_{23} \quad \sigma_{31} \quad \sigma_{12}\}^T \quad (24a)$$

$$\boldsymbol{\varepsilon} = \{\varepsilon_1 \quad \varepsilon_2 \quad \varepsilon_3 \quad \varepsilon_4 \quad \varepsilon_5 \quad \varepsilon_6\}^T = \{\varepsilon_{11} \quad \varepsilon_{22} \quad \varepsilon_{33} \quad \varepsilon_{23} \quad \varepsilon_{31} \quad \varepsilon_{12}\}^T \quad (24b)$$

The initial elastic tensor $\mathbb{C}_0$ can be represented as a six-by-six matrix. For anisotropic materials, it is given as

$$\mathbb{C}_0^{-1} = \left\{ \begin{array}{cccccc} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & -\frac{\nu_{31}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & -\frac{\nu_{32}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{13}}{E_1} & -\frac{\nu_{23}}{E_2} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2G_{23}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2G_{31}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2G_{12}} \end{array} \right\} \quad (25)$$

Since $\nu_{ij}/E_i = \nu_{ji}/E_j$ (i and j are not summation indices), $\mathbb{C}_0$ is symmetric matrix. In this study, the anisotropic effect tensor $\mathbb{M}$ is concerned with the orthonormal basis in the principal direction of the strain, and the anisotropic effect tensor $\mathbb{M}$ can be written as

$$\mathbb{M} = -\frac{3\dot{\varepsilon}_H}{\zeta} \left\{ \begin{array}{cccccc} M_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & M_{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & M_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & (M_{22} + M_{33})/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & (M_{33} + M_{11})/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & (M_{11} + M_{22})/2 \end{array} \right\} \quad (26a)$$

where

$$M_{11} = 3 \left((p_1 \varepsilon_H - q_1)^{-1} + \frac{\sigma_H}{\rho} \right) \dot{\varepsilon}_H + \frac{\dot{\sigma}_H}{\rho} \quad (26b)$$

$$M_{22} = 3 \left((p_2 \varepsilon_H - q_2)^{-1} + \frac{\sigma_H}{\rho} \right) \dot{\varepsilon}_H + \frac{\dot{\sigma}_H}{\rho} \quad (26c)$$

$$M_{33} = 3 \left((p_3 \varepsilon_H - q_3)^{-1} + \frac{\sigma_H}{\rho} \right) \dot{\varepsilon}_H + \frac{\dot{\sigma}_H}{\rho} \quad (26d)$$

When $p_1 = p_2 = p_3 = p$ and $q_1 = q_2 = q_3 = q$, Eq. (26) can be readily reduced to a scalar. Since Eq. (23) may result in asymmetry of the elastic modulus tensor, we derive the evolution equation of the damage model by following a simple symmetrization procedure described in Ref. [55], and presented below as Eq. (27).

$$\dot{\mathbb{C}} = \frac{1}{2} [\mathbb{M} : \mathbb{C}_{AC} + \mathbb{C}_{AC} : \mathbb{M}] \quad (27)$$

A brittle damage model based on tensile brittle failure is presented in this section. However, it should be noted that, for some brittle materials, brittle failure can be induced by tensile loading and compressive loading. Thus, to ensure that both tensile- and compressive-types of brittle fracture are represented, the evolution equation of the proposed damage model is presented as follows:

$$\dot{\mathbb{C}} = \begin{cases} \frac{1}{2} [\mathbb{M}^+ : \mathbb{C}_{AC} + \mathbb{C}_{AC} : \mathbb{M}^+] & \text{for tensile loading} \\ \frac{1}{2} [\mathbb{M}^- : \mathbb{C}_{EF} + \mathbb{C}_{EF} : \mathbb{M}^-] & \text{for compressive loading} \end{cases} \quad (28a)$$

with $\mathbb{C}_{AC}$, $\mathbb{P}_\varepsilon^+$ as in (8b), (8c), and with

$$\mathbb{C}_{EF} = \mathbb{P}_\varepsilon^- : \mathbb{C}_0 : \mathbb{P}_\varepsilon^- \quad (28b)$$

$$\mathbb{P}_\varepsilon^- = \mathbb{I} - \mathbb{P}_\varepsilon^+ \quad (28c)$$

where the $' + '$ and $' - '$ superscripts refer to tensile loading and compressive loading, respectively; $\mathbb{I}$ is the fourth-order identity tensor.

Actual damage growth typically results in failure when the magnitude of the damage variable $\|\mathbb{D}\|$ reaches unity. Therefore, an additional failure criterion can be defined by implementing a critical value for the damage variable D_{cr} [48, 56]. However, because the proposed damage model does not employ the damage tensor $\mathbb{D}$ as its damage variable, the failure criterion for D_{cr} is not adopted. Nevertheless, as failure is observed to occur when the proposed damage value ζ reaches unity, a critical damage value

ζ_{cr} is employed to define the failure criterion in the proposed damage model. Note that the damage value ζ corresponds to a negative value of the strain energy per unit mass (see Eq. (8)).

3. Damage model verification

To validate the proposed brittle damage model, the stress-strain curves obtained via several experiments, which were carried out in previous studies for various polymeric foams (PUF [5, 57–58], PIR [13], R-PUF [7], and polyvinyl chloride (PVC) closed-cell cellular foam [8]), are compared to the numerical simulation results (see Table 1). All tests that are used for comparison in this study (Refs. [5, 7–8, 13, 57–58]) were carried out under the conditions of a quasi-static strain rate and a room temperature environment. To evaluate the orthotropic material properties of polymeric foams, tensile and/or compressive testing was respectively performed for the directions parallel and transverse to the direction that the foam rises, as is shown in Fig. 2.

3.1. Finite element analysis

In the present study, an ABAQUS user-defined subroutine UMAT is developed to perform nonlinear finite element analysis (FEA) on the proposed brittle damage model. UMAT is available to describe the behavior of materials in the absence of an existing material model in the ABAQUS material library. Furthermore, a UMAT subroutine can be used to describe the nonlinear behavior of materials and structures, and conduct damage analysis. However, with this subroutine, the analysis becomes exceedingly computationally expensive when the material behavior is nonlinear. In general, the tangent and secant approaches are used to apply nonlinear solvers [59]. Because using the secant method increases computational costs, the tangent approach is widely used for nonlinear equations. In the present study, we use the algorithmic tangential stiffness (ATS) to describe nonlinear behavior and damage growth [7, 60]

In this study, the proposed brittle damage model is implicitly formulated. The optimal time increment Δt and corresponding strain and stress increments are determined by the Jacobian matrix. By using the hydrostatic stress σ_H^{n+1} , hydrostatic strain ϵ_H^{n+1} , and mass density ρ^{n+1} at the end of the time increment,

and the increment of the hydrostatic stress $\Delta\sigma_H$, hydrostatic strain $\Delta\varepsilon_H$, and damage value $\Delta\zeta$, the anisotropic damage effect tensors are calculated as follows:

$$\mathbb{M}^{\pm,n+1} = \mathbb{M}^{\pm,n+1}(\sigma_H^{n+1}, \Delta\sigma_H, \varepsilon_H^{n+1}, \Delta\varepsilon_H, \rho^{n+1}, \Delta\zeta^{\pm}) \quad (29)$$

where n is the time increment number. The increment of the damage value $\Delta\zeta$ is calculated by solving Eq. (30).

$$\Delta\zeta^+ = \zeta^{+,n+1} - \zeta^{+,n} = \frac{\sum_{i=1}^3 Y_{iikl}^{n+1} C_{AC,klii}^{n+1}}{\rho^{n+1}} - \frac{\sum_{i=1}^3 Y_{iikl}^n C_{AC,klii}^n}{\rho^n} \quad (30a)$$

$$\Delta\zeta^- = \zeta^{-,n+1} - \zeta^{-,n} = \frac{\sum_{i=1}^3 Y_{iikl}^{n+1} C_{EF,klii}^{n+1}}{\rho^{n+1}} - \frac{\sum_{i=1}^3 Y_{iikl}^n C_{EF,klii}^n}{\rho^n} \quad (30b)$$

where

$$Y_{ijkl} = -\frac{1}{2} \varepsilon_{ij} \varepsilon_{kl} \quad (30c)$$

$$C_{AC,ijkl} = P_{\varepsilon,ijmn}^+ C_{0,mnop} P_{\varepsilon,opkl}^+ \quad (30d)$$

$$C_{EF,ijkl} = P_{\varepsilon,ijmn}^- C_{0,mnop} P_{\varepsilon,opkl}^- \quad (30e)$$

$$P_{\varepsilon,ijkl}^- = I_{ijkl} - P_{\varepsilon,ijkl}^+ \quad (30f)$$

$$\rho = \rho_0 \exp(-3\varepsilon_H) \quad (30g)$$

Then, the increment of the elastic modulus tensor is determined as follows:

$$\Delta C_{ijkl} = \begin{cases} \frac{1}{2} [M_{ijmn}^{+,n+1} C_{AC,mnkl}^{n+1} + C_{AC,ijmn}^{n+1} M_{mnkl}^{+,n+1}] & \text{for tensile loading} \\ \frac{1}{2} [M_{ijmn}^{-,n+1} C_{EF,mnkl}^{n+1} + C_{EF,ijmn}^{n+1} M_{mnkl}^{-,n+1}] & \text{for compressive loading} \end{cases} \quad (31)$$

The updated stress and elastic modulus tensor yield

$$\sigma_{ij}^{n+1} = \sigma_{ij}^n + C_{ijkl}^{n+1} \Delta\varepsilon_{kl} + \Delta C_{ijkl} \varepsilon_{kl}^{n+1} \quad (32a)$$

$$C_{ijkl}^{n+1} = C_{ijkl}^n + \Delta C_{ijkl} \quad (32b)$$

To obtain the updated hydrostatic stress $\sigma_H^{n+1} (= \sigma_H^n + \Delta\sigma_H)$, the increment of the hydrostatic stress $\Delta\sigma_H$ in Eq. (32) is rearranged as

$$\begin{aligned} \Delta\sigma_H & \left[1 + \sum_{j=1}^3 \sum_{i=1}^3 \left(\frac{\Delta\varepsilon_H (1+3\Delta\varepsilon_H)}{\rho^{n+1}} \frac{C_{AC,iijj}^{n+1}}{\Delta\zeta^+} (\Delta\varepsilon_{jj} + \varepsilon_{jj}^{n+1}) \right) \right] \\ & = \sum_{j=1}^3 \sum_{i=1}^3 \left(\frac{C_{iijj}^n \Delta\varepsilon_{jj}}{3} - \Delta\varepsilon_H \frac{\bar{M}_i^+ + \bar{M}_j^+}{2} \frac{C_{AC,iijj}^{n+1}}{\Delta\zeta^+} (\Delta\varepsilon_{jj} + \varepsilon_{jj}^{n+1}) \right) \text{ for tensile loading} \end{aligned} \quad (33a)$$

$$\begin{aligned}
& \Delta\sigma_H \left[1 + \sum_{j=1}^3 \sum_{i=1}^3 \left(\frac{\Delta\varepsilon_H(1+3\Delta\varepsilon_H)}{\rho^{n+1}} \frac{C_{EF,ijj}^{n+1}}{\Delta\zeta^-} (\Delta\varepsilon_{jj} + \varepsilon_{jj}^{n+1}) \right) \right] \\
& = \sum_{j=1}^3 \sum_{i=1}^3 \left(\frac{C_{ijj}^n \Delta\varepsilon_{jj}}{3} - \Delta\varepsilon_H \frac{\tilde{M}_i^- + \tilde{M}_j^-}{2} \frac{C_{EF,ijj}^{n+1}}{\Delta\zeta^-} (\Delta\varepsilon_{jj} + \varepsilon_{jj}^{n+1}) \right) \text{ for compressive loading}
\end{aligned} \tag{33b}$$

where

$$\tilde{M}_i = 3 \left((p_i \varepsilon_H^{n+1} - q_i)^{-1} + \frac{\sigma_H^n}{\rho^{n+1}} \right) \Delta\varepsilon_H \tag{33c}$$

The ATS tensor C_{ijkl}^{ATS} is defined here as follows:

$$C_{ijkl}^{ATS} = \frac{\partial \Delta\sigma_{ij}}{\partial \Delta\varepsilon_{kl}} = C_{ijkl}^{n+1} + \frac{\Delta C_{ijmn} \varepsilon_{mn}^{n+1}}{\Delta\varepsilon_{kl}} \tag{34}$$

After the n -th iteration concludes, the $(n + 1)$ -th iteration begins as a new increment of time. A single iteration, in which all incremental values are updated, is carried out by defining the solution-dependent state variables (SDVs). The SDVs are initialized by ABAQUS using the subroutine SDVINI.

The proper material constants p and q of the proposed model are determined based on the effects of the material constants on the damage growth (see Section 3.2 and 3.3). Since the initial decrease in the elastic modulus decreases as q increases, the material constant q is first determined, following which p is found based on the experimental results.

The damage evolution can be modeled as the elimination of elements in the finite element mesh. The elimination of elements can be realized via two methods: the element elimination and element weakening method (EWM) [61–62]. In the present simulation, EWM is used to describe the failure behavior [63–64]. The principle of this approach is that when the damage value ζ in a certain finite element is larger than or equal to critical value ζ_{cr} (see Section 3.4), failure occurs and the elastic stiffness of this element is set to a very low value (50 Pa, i.e., about 0.00001 % of the initial value) [63]. In addition, to avoid numerical problems of strong local loss of equilibrium, the stress in relaxation steps is set to zero [64]. For clarity, EWM-based proposed damage model numerical algorithm in UMAT is illustrated in Algorithm 1.

3.2. Tensile tests

In this study, FEA is carried out using only one 8-node element (C3D8) for finite element identification [65] because there was no information on the dimensions of the specimens in previous studies. For the boundary conditions, one side of the element is fixed, while displacement control is implemented on the other side. For the coordinate system, the foaming direction is defined as the z-axis direction, and the transverse directions correspond to the x- and y-axes directions.

Fig. 3 shows the experimental and simulated stress-strain relations under tensile loading for R-PUF and PUF (density: 128.0 kg/m^3), respectively. Table 2 lists the respective material properties, constants, and critical damage values for the R-PUF and PUF (density: 128.0 kg/m^3) implemented in the proposed damage model. As can be observed in Fig. 3, the constitutive and evolution equations comprising the proposed model describe the brittle material behavior of each material with sufficient accuracy, as the simulation results are in good agreement with the experimental results. In the simulations, the brittle failure point of PUF is precisely predicted; however, there are some discrepancies between the simulated and experimental failure points of R-PUF because R-PUF is not a perfectly porous material. As the directions of the R-PUF fibers are parallel to the transverse directions, the strength along the transverse directions is higher than that of non-fibrous materials. This explains why the brittle failure point along the transverse direction is found to be lower in the simulations than in the experiments. However, with the exception of the failure point results, the simulation results are shown to adequately describe the material behavior of R-PUF.

To demonstrate the effects of the material constants p and q on damage growth, three additional p and q values are respectively simulated for further analysis of tensile testing for the rise direction on PUF (density: 128.0 kg/m^3). As is shown in Fig. 4a, the decrement of the secant modulus of elasticity increases in response to an increase in p ; conversely, the initial decrement of the secant modulus of elasticity decreases as q increases until the change in the secant modulus of elasticity becomes positive as q increases (see Fig. 4b). Thus, it is important to implement an appropriate value for the material constant q such that the increment of the secant modulus of elasticity is negative. Based on the respective relationships between material constants and damage growth, both $1/p$ and q can describe the resistance to the growth and coalescence of cavities.

3.3. Compressive tests

Generally, a typical compressive stress-strain curve for polymeric foams comprises three defined regimes: elastic, plateau, and densification (see Fig. 1b) [11, 15, 66–69]. In an elastic regime, cell edge bending or face stretching occurs [66]. In the plateau regime, the stress rarely increases in response to an increased load, and the foam cell starts to collapse by elastic buckling, plastic yielding or brittle crushing, depending on the mechanical properties of the cell wall [67]. In the densification regime, the stress rapidly increases as the load increases [68], causing cells to collapse throughout the material [69].

In the plateau region, a shear crack is initiated [70] and wrinkles and micro-cracks evolve into cracks with increases in the strain [71], and cracks are initiated in the plateau regime. Because the cracks indicate the onset of the initial failure, the proposed brittle damage model only describes the elastic regime under compressive loading. Figs. 5 and 6 show the experimental and simulated stress-strain relationships under compressive loading for PVC and PIR, respectively. The material properties, constants, and critical damage values are listed in Table 3 for the proposed PVC and PIR damage models. As can be seen, each of the proposed models describes the respective material behavior in the elastic regime with sufficient accuracy. Moreover, the simulation results show that the PVC and PIR elastic failure points are precisely predicted.

In order to demonstrate the effects of the material constants p and q on damage growth, three hypothetical p and q values are implemented to simulate compressive testing for the transverse direction on the proposed PIR damage model. Fig. 7 shows that the decrement of the secant modulus of elasticity increases with an increase in the absolute value of p , whereas the initial decrement of the secant modulus of elasticity decreases as q increases. Under compression, the effects of the material constants on damage initiation and growth are similar to those observed in the application of tensile loads (see Section 3.2).

3.4. Failure criterion for the proposed damage model

In this study, the failure energy density E_f is calculated for various polymeric foams by referring to the literature to determine the corresponding anisotropic mechanical properties for the polymeric foams implemented in the proposed brittle damage model (see Table 1). Mass density is generally thought to influence the failure energy density since mass density is related to the void volume fraction [50], which describes the state of damage; Fig. 8 shows the relationship between mass density and failure energy density. Although there are few tensile testing data points because of a lack of experimental results for orthotropic polymeric foams, the results indicate that the failure energy density increases as mass density increases.

Power law regression analysis is carried out to determine an empirical formula that can accurately quantify the relationship between the mass density and failure energy per unit mass. The empirical formula is presented as follows:

$$\zeta_{cr} = -a \cdot \rho_0^b \quad (35)$$

where a and b are the empirical formula variables (see Table. 4). This formula is used to determine the failure point of the proposed brittle damage model.

4. Concluding remarks

In the current study, a novel brittle damage model is developed as based on the free energy imbalance inequality for polymeric foams comprising porous materials and voids. The developed damage model describes the process of growth and coalescence of voids by quantifying the effects of hydrostatic stress and strain and mass density on damage initiation and growth. Furthermore, in the developed anisotropic brittle damage model, the mechanical effects are dependent on the sign of the strain value; more explicitly, the proposed model is able to represent the unilateral effect.

The proposed brittle damage model is implemented in the ABAQUS subroutine UMAT to perform nonlinear FEA. The simulated results of the proposed model are in agreement with the nonlinear behavior of polymeric foams. Furthermore, the effects of material constants on damage growth are evaluated.

In general, material failure occurs when the damage value reaches the value of the critical damage variable D_{cr} . However, because the proposed damage model cannot implement D_{cr} , a failure energy per

unit mass ζ_{cr} is used to determine whether material failure has occurred. The empirical formula defining the relationship between the failure energy per unit mass and mass density is determined by performing power law regression analysis on the simulated results. By consequence, the failure point of the polymeric foam is determined as based on the empirical formula.

The developed brittle damage model possesses the following technical merits and demerits for the field of continuum mechanics. The developed brittle damage model is deemed as suitable for polymeric foams because the damage model is shown to accurately represent the effects of voids on damage growth damage propagation and material failure. However, the material behavior cannot be accurately predicted for a fiber-reinforced composite using this damage model because the effects of brittle damage along the fiber direction are disregarded. Despite this disadvantage, as it can be used as a reference for the development of a modified damage model that considers the effects of fibers, the developed model is expected to be useful for damage analysis of the polymeric foams commonly applied in the field of continuum mechanics.

In the future, the following points must be considered for further development of the current model:

1. Localization of deformation in polymeric foams under tensile loading using the damage model.
2. Verification of the damage model and failure criterion via multiaxial experiments with nonproportional loading.
3. Discussion on the behavior of the damage surface under multi-axial loading conditions.
4. Development of a theory accounting for damage growth due to cyclic loading.
5. Implementation of the proposed model in the finite element analysis for complex loading states with non-homogeneous stress states
6. Extension of the proposed model for the description of the plateau regime and hysteresis behavior under compressive loading

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Algorithm 1 Computational algorithm for ABAQUS UMAT.

Input: $\Delta\epsilon$, σ^n , ϵ^n , $\mathbb{C}^n$

Output: σ^{n+1} , ϵ^{n+1} , $\mathbb{C}^{n+1}$

(1) *Updated strain*

$$\epsilon^{n+1} = \epsilon^n + \Delta\epsilon^n$$

(2) *Damage value*

$$\mathbb{Y}^{n+1} = -\frac{1}{2}\epsilon^{n+1} \otimes \epsilon^{n+1}$$

$$\rho^{n+1} = \rho_0 \exp[-tr(\epsilon^{n+1})]$$

$$\text{Update } \zeta^{n+1} = tr(\mathbb{Y}^{n+1}/\rho^{n+1}; \mathbb{P}_{\epsilon^{n+1}}: \mathbb{C}_0: \mathbb{P}_{\epsilon^{n+1}})$$

$$\Delta\zeta = \zeta^{n+1} - \zeta^n$$

(3) *Increment of elastic modulus tensor*

$$\text{Update } \sigma_H^{n+1} (= \sigma_H^n + \Delta\sigma_H) \text{ by solving } \sigma^{n+1} = \sigma^n + \mathbb{C}^{n+1}: \Delta\epsilon + \Delta\mathbb{C}: \epsilon^{n+1}$$

$$\text{where } \Delta\mathbb{C} = \frac{1}{2} [\mathbb{M}^{n+1}: \mathbb{P}_{\epsilon^{n+1}}: \mathbb{C}_0: \mathbb{P}_{\epsilon^{n+1}} + \mathbb{P}_{\epsilon^{n+1}}: \mathbb{C}_0: \mathbb{P}_{\epsilon^{n+1}}: \mathbb{M}^{n+1}]$$

Calculate $\mathbb{M}$ and $\Delta\mathbb{C}$ using Eq. (29) and Eq. (32), respectively

(4) *Updated stress and failure criterion*

if $|\zeta^n| \geq |\zeta_{cr}|$ **then**

| Update $\mathbb{C}^{n+1} = 50 Pa$

| $\sigma^{n+1} = \mathbf{0}$

else

| Update $\mathbb{C}^{n+1} = \mathbb{C}^n + \Delta\mathbb{C}^n$

| $\sigma^{n+1} = \sigma^n + \mathbb{C}^{n+1}: \Delta\epsilon + \Delta\mathbb{C}: \epsilon^{n+1}$

end

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Table 4 Regression variables.

Table 1 Studies on the anisotropic mechanical properties of polymeric foams.

Literature	Target material	Load type	Mass density
Lee and Lee (2014)	R-PUF	Tension	125.0 kg/m ³
Daniel and Cho (2011)	PVC	Tension & Compression	250.0 kg/m ³
Hamilton et al. (2013)	PUF	Tension & Compression	128.0 kg/m ³
Huber and Gibson (1988)	PUF	Compression	96.0 kg/m ³
Huber and Gibson (1988)	PUF	Compression	28.0 kg/m ³
Tu et al. (2001)	PUF	Compression	25.6 kg/m ³
Anderson et al. (2016)	PIR	Compression	34.0 kg/m ³

Table 2 Material properties, constants, and critical damage values for the proposed brittle damage model of R-PUF and PUF subjected to tensile loading.

	R-PUF	PUF
E_t (MPa)	195.00	45.00
E_r (MPa)	100.00	75.00
G_t (MPa)	77.00	20.00
G_r (MPa)	85.47	20.45
ν_{tt}	0.17	0.10
ν_{tr}	0.34	0.20
p_t (s^2/mm^2)	2.40E-10	2.70E-10
p_r (s^2/mm^2)	3.15E-10	3.30E-10
q_t (s^2/mm^2)	1.00E-12	6.00E-12
q_r (s^2/mm^2)	2.60E-12	1.70E-12
ζ_{cr} (mm^2/s^2)	-2.00E+08	-5.35E+08

Note: Here, ' t ' and ' r ' subscripts denote the transverse and rise directions, respectively.

Table 3 Material properties, constants, and critical damage values for the proposed brittle damage models of PVC and PIR subjected to compressive loading.

	PVC	PIR
E_t (MPa)	201.00	2.50
E_r (MPa)	322.00	9.50
G_t (MPa)	85.00	2.00
G_r (MPa)	75.56	1.09
v_{tt}	0.33	0.15
v_{tr}	0.20	0.10
p_t (s^2/mm^2)	-3.10E-10	-9.00E-10
p_r (s^2/mm^2)	-2.27E-10	-1.03E-09
q_t (s^2/mm^2)	1.90E-12	3.10E-11
q_r (s^2/mm^2)	8.70E-13	2.50E-12
ζ_{cr} (mm^2/s^2)	-5.75E+08	-1.30E+08

Note: Here, ' t ' and ' r ' subscripts denote the transverse and rise directions, respectively.

Table 4 Regression variables.

Variables	a	b	R^2
Tensile loading	0.0006	2.7428	0.8463
Compressive loading	11.093	0.8415	0.6173

Note: Here, R^2 is the coefficient of determination for the empirical formula.

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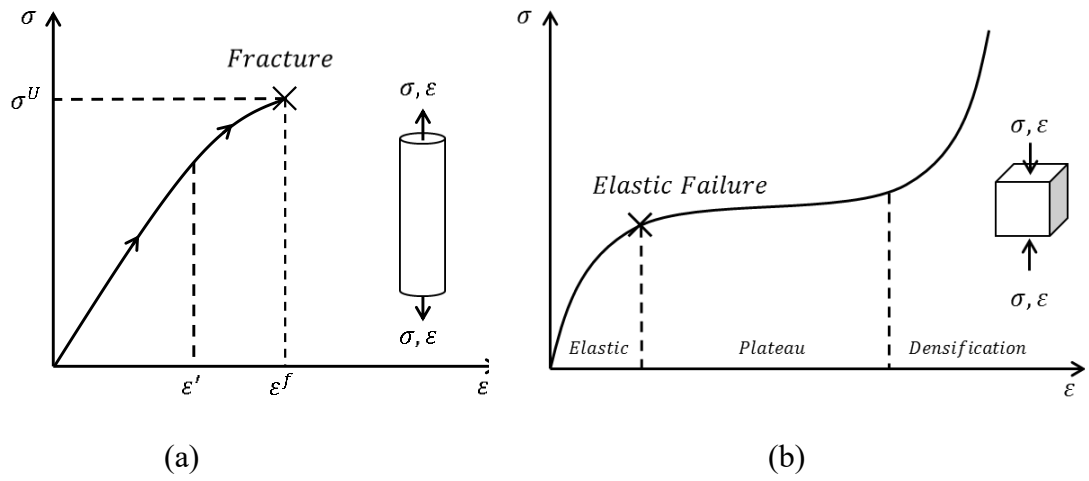


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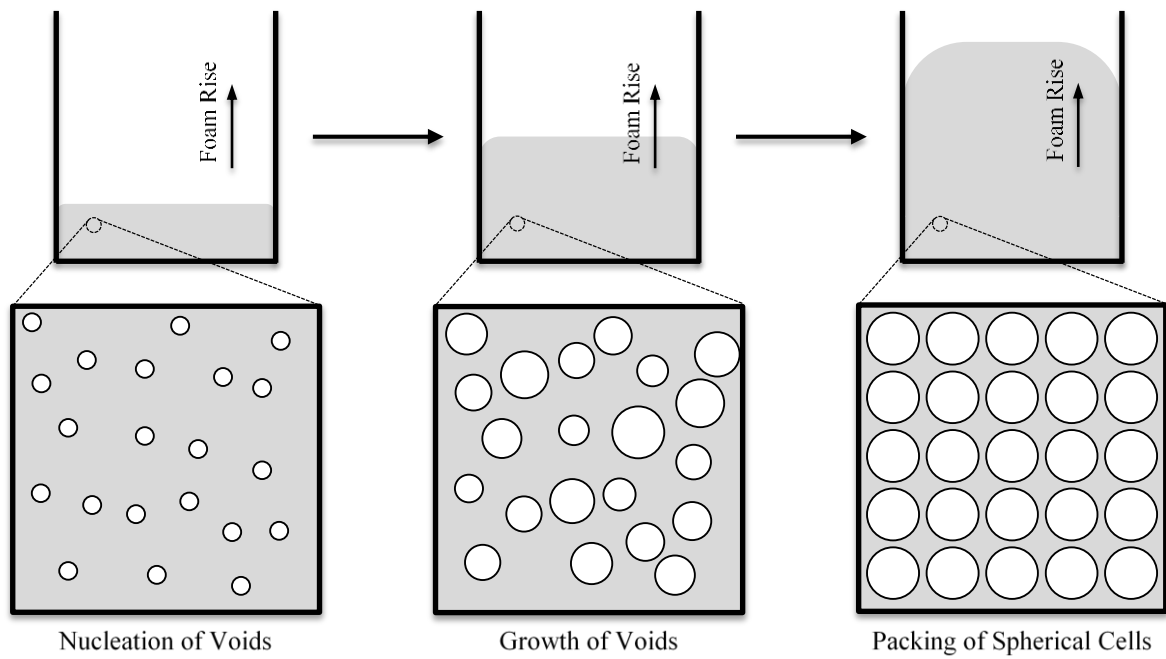


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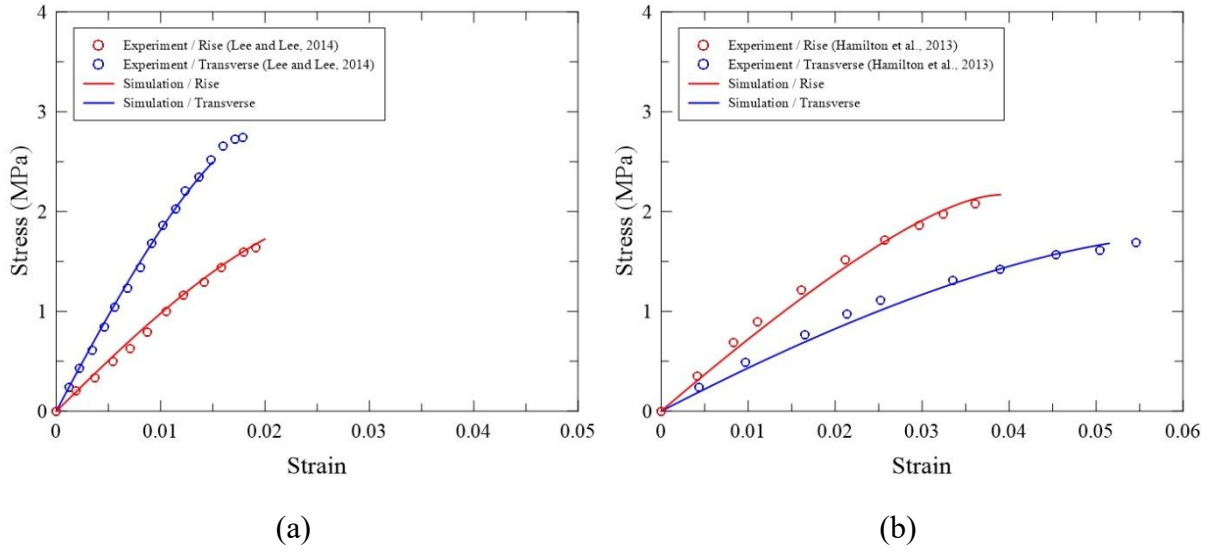


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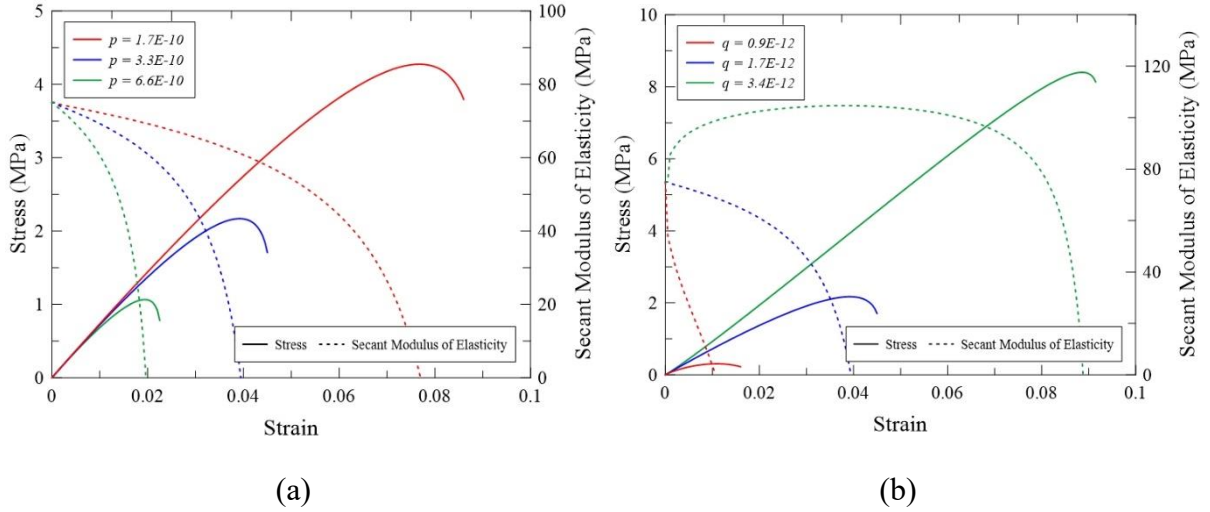


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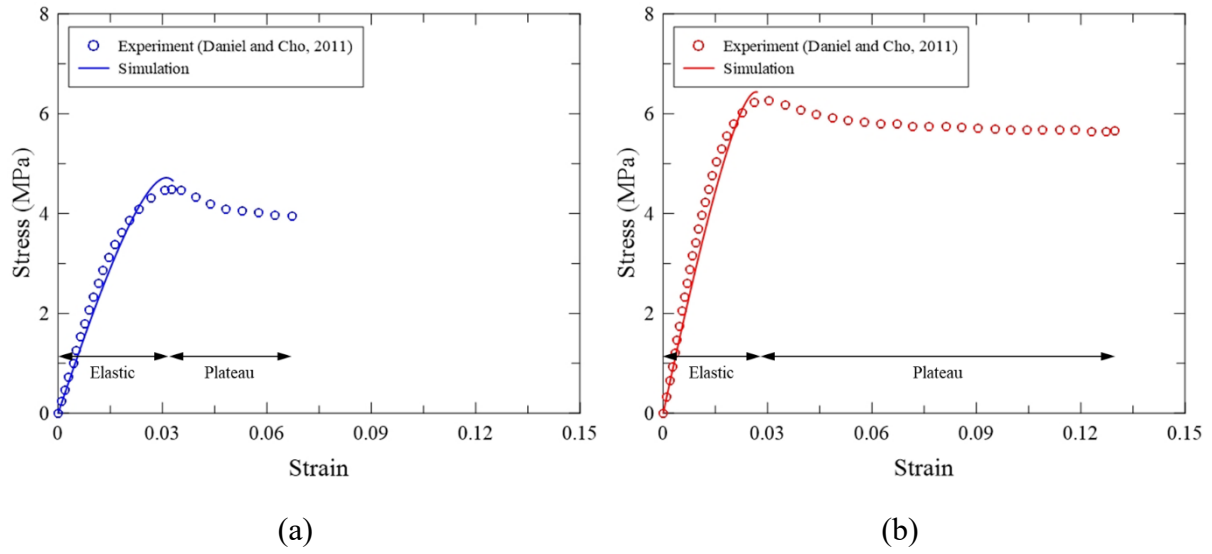


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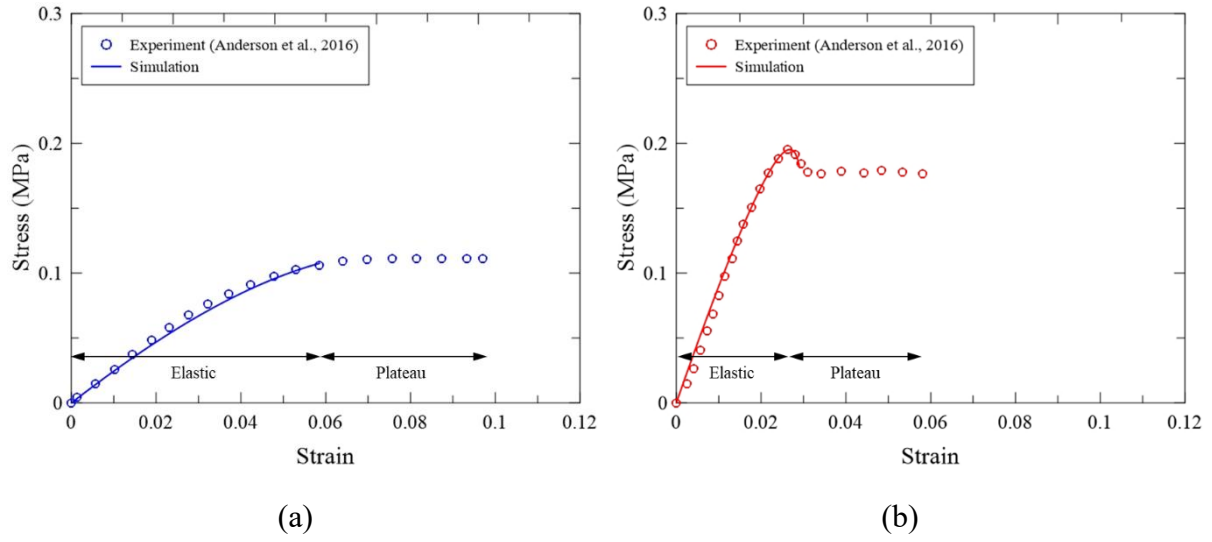


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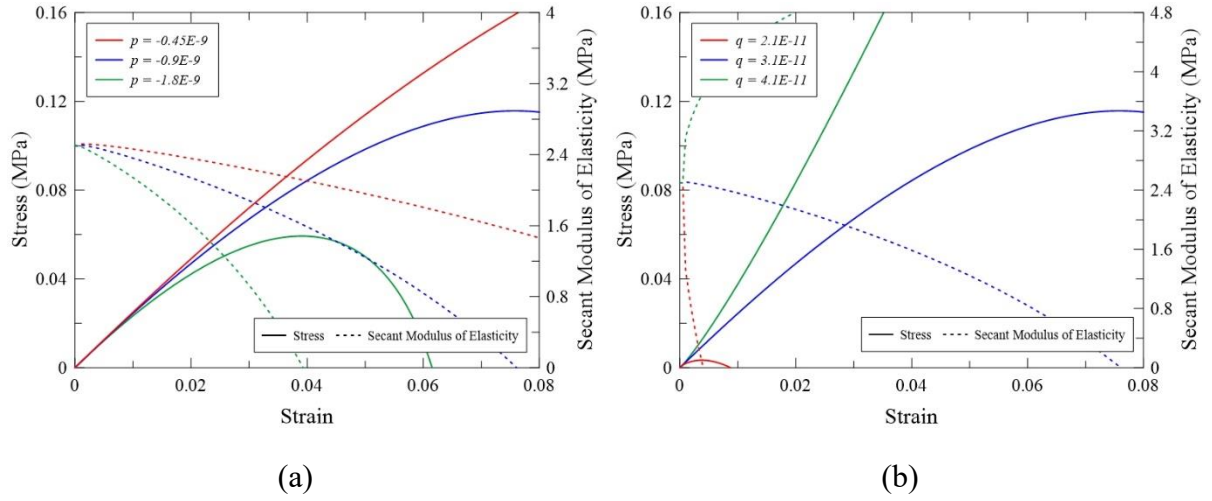


Fig. 7. Demonstration of the effects of (a) p and (b) q on damage growth. The solid and dotted lines correspond to the stress-strain curves and secant modulus of elasticity, respectively. For each material constant, three additional values are simulated for analysis of PIR compression testing for the transverse direction.

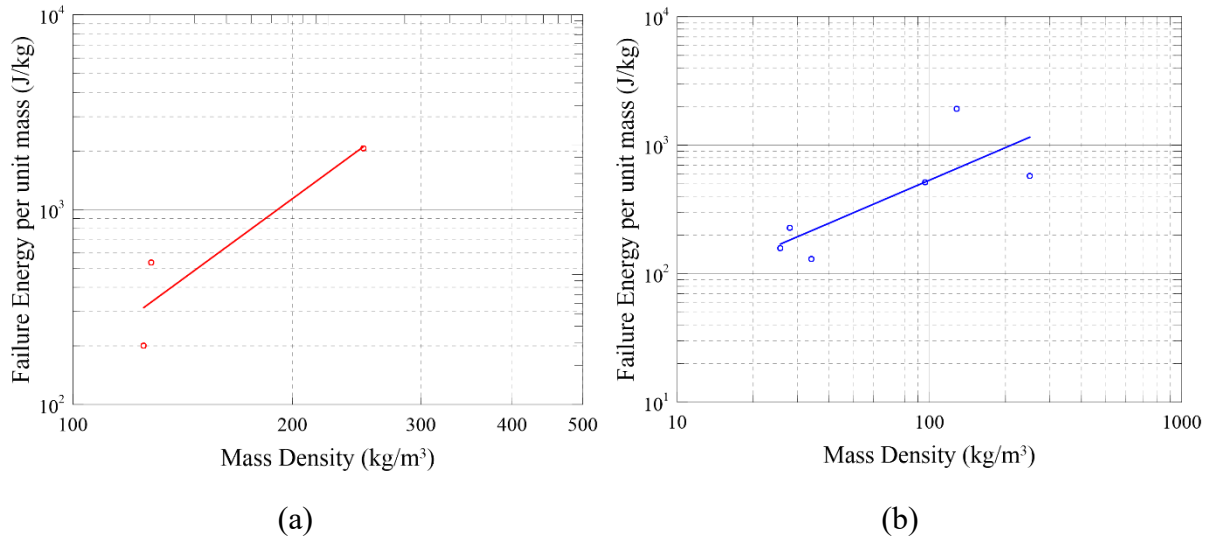


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